

RESET-FREE SWING-UP AND HOLD OF SEVEN THROUGH TEN SERIAL LINKS WITH ONE CART ACTUATOR

Funnel Composition, Target-Plant Trajectory Optimization, and Evidence-Gated Scaling

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Technical report, revision 2
14 September 2026
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EVIDENCE SNAPSHOT

LINKS	NOISY GATES	HELD-OUT HOLD	MAX CART
7	20/20 + 100/100	15.48 s	2.3708 m
8	20/20 + 100/100	12.22 s	2.9300 m
9	20/20 + 100/100	12.18 s	2.9800 m
10	20/20 + 100/100	6.12 s	1.9976 m

PROVISIONAL PUBLIC LINK-COUNT RECORD

Scoped to the frozen simulated benchmark in this report. Independent reproduction and matched prior-art adjudication remain open.

Abstract

We report reset-free swing-up and sustained upright stabilization of uniform serial cart-poles with seven, eight, nine, and ten passive links, driven by one bounded horizontal cart force. Every plant starts from a noisy hanging state and uses the same three-phase control architecture in one uninterrupted MuJoCo episode: a hanging-equilibrium LQR contracts the launch distribution, a target-plant Box-FDDP trajectory with saved time-varying feedback performs the swing, and an upright LQR captures and maintains the chain. The plant has total link length 3 m, total link mass 1 kg, cart mass 1 kg, force limit ± 80 N, rail limit ± 3 m, and a 50 Hz control rate. Success requires every absolute link angle to remain within 0.15 rad of upright continuously for at least 5 s.

For each link count, disjoint noisy 20-episode and 100-episode cohorts passed without a rail failure or state reset: 480/480 noisy trials in total. The eight-, nine-, and ten-link releases also passed 60/60 exact-start checks. Four held-out 30 s videos use seeds outside the noisy cohorts. The ten-link run reaches upright at 23.90 s and holds for the remaining 6.12 s. The sequence exposes a reusable control principle: do not ask a narrow global swing controller to absorb all uncertainty. First use an inexpensive stable-equilibrium controller to contract the initial distribution, then optimize the nonlinear route to enter a verified local capture funnel quietly.

A structured public search performed on 2026-09-14 found no directly comparable seven-or-more-link cart-actuated swing-up-and-hold result with a frozen plant, bounded rail and force, noisy hanging starts, sustained all-link upright gate, public controller, and multi-seed evidence. We therefore claim a provisional public link-count record for this precisely declared simulated benchmark. It is not a universal world record across unmatched plants or rules, and independent reproduction remains open.

1. Contributions

This report makes five contributions.

1. It consolidates four separately frozen results into one benchmark and evidence contract, spanning seven through ten serial links.
2. It identifies settled-launch funnel composition as the common mechanism that made narrow target-plant trajectories robust to the declared hanging-start distribution.
3. It isolates the count-dependent changes that were actually required: cart parking, terminal angle/rate shaping, longer arrival shaping, and a longer ten-link route. The controller architecture did not change.
4. It publishes disjoint noisy gates, exact checks, held-out videos, SHA-256 manifests, negative controls, and replay commands instead of relying on one hand-selected trajectory.
5. It states a falsifiable record claim tied to a precise benchmark, while separating that claim from ongoing arbitrary-morphology solver research.

2. Problem definition

2.1 Dynamics and underactuation

Let the generalized coordinate be

$$\mathbf{q} = [x, \theta_1, \dots, \theta_n]^T,$$

where x is cart position and the θ_i are relative revolute-joint angles. The serial-chain dynamics have the standard manipulator form

$$\begin{aligned} M(\mathbf{q}) \ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) + \mathbf{d}(\dot{\mathbf{q}}) &= \mathbf{B} \mathbf{F}, \\ \mathbf{B} &= [1, 0, \dots, 0]^T, \quad |\mathbf{F}| \leq 80 \text{ N}. \end{aligned}$$

There are $n+1$ generalized coordinates but only one actuator. The absolute orientation of link i is

$$\alpha_i = \sum(\theta_j, j=1..i).$$

The success gate is evaluated on the absolute angles, not on the relative joint coordinates:

$\max_i |\text{wrap}(\alpha_i)| < 0.15$ rad continuously for at least 5.0 s.

This matters increasingly with link count. Small relative-angle errors can accumulate into a large distal-link error, and low relative hinge-rate RMS does not guarantee that every physical link has low absolute angular velocity.

2.2 Frozen benchmark

All four released plants use the same aggregate contract.

Quantity	Frozen value
Passive serial links	7, 8, 9, or 10
Total chain length	3.0 m
Total chain mass	1.0 kg
Cart mass	1.0 kg
Cart-center rail limit	+/-3.0 m
Cart force	continuous, +/-80 N
MuJoCo integration step	0.005 s
Control frame skip	4
Policy rate	50 Hz
Episode duration	30.0 s
Start	hanging, with Gaussian perturbations
Angle and generalized-velocity noise	standard deviation 0.05
Upright set	all absolute angles within 0.15 rad
Required continuous hold	5.0 s
State resets after episode start	prohibited

The links are uniform within each plant: length $3/n$ m and mass $1/n$ kg. The initial relative-angle vector is $[\pi, 0, \dots, 0]$ plus noise. An episode receives one seeded reset at its beginning. Phase transitions carry forward position, velocity, simulation time, and all hidden simulator state unchanged.

2.3 Dimensionless view

For total chain length L , total moving mass M , and gravity g , define the natural time $t_0 = \sqrt{L/g}$. The main scale-free groups are force authority $F_{\max}/(Mg)$, rail ratio R/L , control period dt/t_0 , cart-to-chain mass ratio, link length fractions, link mass fractions, and dimensionless damping and armature. These groups support analytic transfer between physically similar plants, but they do not make one finite-horizon force trace universal across link counts. Internal modal structure still changes with n ; every released route is therefore refined and gated on its exact target plant.

3. One controller architecture for all four plants

The runtime policy is a composition of three feedback funnels:

```

noisy hanging distribution
  |
  v
hanging LQR contraction and cart parking
  |
  v
Box-FDDP swing route with time-varying feedback
  |
  v
quiet terminal state inside the upright LQR basin

```

|
v

sustained all-link upright hold

3.1 Phase A: contract the launch distribution

The hanging configuration is stable and inexpensive to regulate. A finite- difference linearization at that equilibrium produces a discrete LQR that damps the reset perturbations and steers the cart to a chosen park target. If x_p is that target and x_h is the measured state, the phase uses

$$u_h = \text{clip}(-K_h (x_h - x_h^*), -1, 1).$$

This is active control, not waiting and not a simulator reset. Its purpose is to map a relatively broad initial distribution into the narrow launch basin of the nonlinear swing route. The seven-link release centers the cart. Links 8-10 use a small negative park target to reserve rail room in the direction required by the selected swing branch.

3.2 Phase B: track a target-plant nonlinear swing route

Box-FDDP optimizes bounded normalized actions on the exact MuJoCo target plant. The controller stores nominal actions $u_{\text{bar}}[k]$, nominal states $z_{\text{bar}}[k]$, and time-varying gains $K[k]$. Execution uses

$$u[k] = \text{clip}(u_{\text{bar}}[k] + \text{kappa } K[k] \text{ e_periodic}(z[k], z_{\text{bar}}[k]), -1, 1),$$

where z contains dimensionless cart states and absolute link orientations. `e_periodic` evaluates revolute-joint errors on the nearest angular branch. The nominal cart channel is translated to the measured parked cart position, an exact symmetry of the horizontal plant away from rail contact.

The route objective does more than cross upright. It must deliver low absolute angle error, low internal rate, modest cart velocity, and feasible rail position at the route boundary. A visually upright crossing with unresolved internal momentum is not a valid handoff.

3.3 Phase C: capture and maintain

At the route horizon, an upright linearization supplies a saturated LQR around the same cart target. The switch is delayed until the planned route boundary. The local expert is intentionally not responsible for creating swing energy or arresting a high-energy chain. The nonlinear route must enter its practical capture basin first.

3.4 Why the composition works

Let D_0 be the noisy hanging distribution, D_s the contracted launch set, and F_u the practical upright capture funnel. The design objective is the set map

$$D_0 \xrightarrow{\text{--pi_settle--}} D_s \xrightarrow{\text{--pi_swing--}} F_u \xrightarrow{\text{--pi_LQR--}} x_{\text{upright}}.$$

The principal insight is organizational. Robustness is easier when assigned to the controller phase that can obtain it cheaply. The stable hanging LQR reduces launch uncertainty; trajectory feedback corrects along the global maneuver; the upright LQR handles only local maintenance. Trying to make any one phase do all three jobs produced the strongest negative controls in this project.

4. How the solution scaled from seven to ten links

The same architecture survived every promotion. What changed was the target- plant route and a small number of measured phase parameters.

Links	Hanging phase	Park target	Swing route	Route feedback	Terminal shaping
7	10.0 s	0.00 m	228 steps / 4.56 s	2.00	quiet terminal state
8	14.0 s	-0.15 m	197 steps / 3.94 s	0.75	quiet terminal state
9	14.0 s	-0.05 m	197 steps / 3.94 s	1.00	angle 20x, hinge rate 4x
10	16.0 s	-0.05 m	400 steps / 8.00 s	1.00	angle 20x, hinge rate 4x

4.1 Seven links: correct coordinates and a settled launch

The original exact-state route was excellent but failed the noisy gate 0/20. The decisive repair had two parts. First, optimizer warm-start states were rebuilt from MuJoCo position and velocity because an earlier artifact confused cumulative absolute rates with relative hinge rates. Second, a 10 s hanging LQR contracted the reset distribution before launching the 4.56 s route. Translating the nominal cart coordinate to the measured settled location and using route feedback scale 2.0 produced 20/20 and 100/100.

4.2 Eight links: park the cart to use the rail asymmetrically

Direct seven-to-eight transfer failed. The exact eight-link route retained the three phases, but its centered swing consumed slightly too much positive rail. A 14 s park at -0.15 m used translational symmetry to shift the route without changing link dynamics. Open-loop replay could pass an exact start but scored 0/20 with noise; saved route feedback was essential. A shorter 12 s park at -0.20 m reached 95/100 and was rejected in favor of the frozen 100/100 setting.

4.3 Nine links: optimize for the capture state, not the crossing

The first nine-link route arrived near upright with maximum angle about 0.314 rad, outside a useful local basin. Increasing the terminal absolute-angle block by 20x and the hinge-rate block by 4x produced a route-boundary state with maximum absolute angle about 0.00004 rad, hinge-rate RMS about 0.00068 rad/s, and cart velocity about 0.00041 m/s. A 14 s park at -0.05 m then left enough canonical rail margin for the 3.94 s feedback route.

4.4 Ten links: shape an arrival segment, then re-optimize feedback

The ten-link campaign exposed the difference between one quiet sample and a robust arrival. Several routes ended almost exactly upright yet required impossible saturated feedback immediately afterward. The successful warm start first shaped the final two seconds of arrival. Box-FDDP then re-optimized an 8 s target-plant route and its feedback with a 2.85 m soft rail, inside the actual 3.0 m benchmark rail. A 16 s park reduced launch residuals. This route enters the local basin with enough time to complete a 6.12 s hold before the 30 s limit.

5. Evaluation protocol

5.1 Promotion gates

Each release had to satisfy all of the following:

1. A 20-episode noisy hanging-start gate.
2. A disjoint 100-episode noisy hanging-start gate.
3. Zero state resets after the initial seeded reset.
4. No cart-center rail violations.
5. At least 5 continuous seconds with every absolute link angle inside 0.15 rad.
6. A 30 s held-out video with a seed outside both noisy cohorts.
7. A manifest binding the config, generated XML, controller, gates, video, and

metadata by SHA-256.

Links 8-10 additionally have 20-episode zero-noise checks. Exact checks are diagnostics, not substitutes for noisy trials.

5.2 Statistical interpretation

Every link count passed 100/100 in its larger noisy cohort. A two-sided 95% Clopper-Pearson interval for 100 successes in 100 Bernoulli trials has a lower bound of approximately 0.964. This does not prove certainty or robustness outside the declared distribution. Pooling all four counts into 400/400 would also be misleading because the plants and controllers differ. The correct statement is four separate 100/100 results under four frozen, closely related benchmarks.

5.3 Independent evidence layers

The evidence stack deliberately separates planning, execution, and presentation:

- planner artifacts contain the target trajectory and local feedback policy;
- evaluation artifacts execute that policy in the exact target MuJoCo plant;
- release verifiers check hashes, cohorts, dimensions, outcomes, and metadata;
- videos are state-faithful renders of held-out simulated trajectories;
- negative controls remain public and are not relabeled as successful runs.

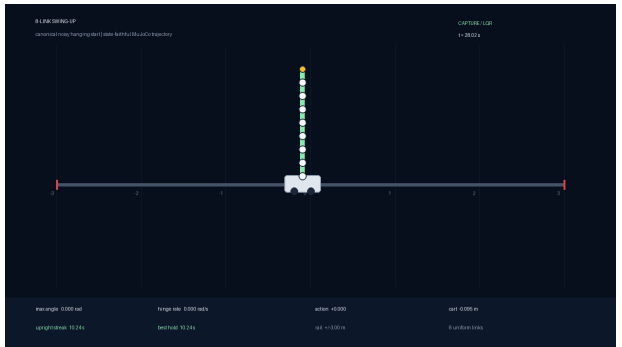
6. Results

6.1 Main outcomes

Links	Noisy 20	Noisy 100	Exact 20	First upright in video	Hold through video end	Max cart, noisy 100
7	20/20	100/100	not a release gate	14.54 s	15.48 s	2.3708 m
8	20/20	100/100	20/20	17.80 s	12.22 s	2.9300 m
9	20/20	100/100	20/20	17.84 s	12.18 s	2.9800 m
10	20/20	100/100	20/20	23.90 s	6.12 s	1.9976 m



7 links at t=28.02 s



8 links at t=28.02 s



9 links at t=28.02 s



10 links at t=29.98 s

Figure 1. Held-out state-faithful video frames. All links are upright in one uninterrupted episode; the ten-link frame is taken near the 30 s boundary after the five-second gate has passed.

All 400 episodes in the four 100-episode noisy gates reached the 30 s time limit. None ended at the rail. Across the separate noisy 20-episode cohorts, another 80/80 passed. The 60 exact checks for links 8-10 also passed.

6.2 Rail use is a route property

The nine-link result approaches the 3 m cart-center boundary most closely, at 2.9800 m. Ten links uses substantially less rail despite having more internal modes, because its longer route and constrained arrival were optimized as a different branch. Link count alone therefore does not determine rail demand. For a trajectory with cart positions x_k , cart half-length r_c , and desired clearance c , the body-aware requirement is

$$\text{rho_required} = (\max_k |x_k| + r_c + c) / L.$$

This quantity should be reported alongside force authority, timing, damping, and route construction. A single empirical line from link count to rail length would hide the dominant controller dependence visible in the table above.

6.3 What the results establish

The evidence establishes that one phase architecture can swing up and hold four increasingly underactuated uniform serial chains under the declared simulation contract. It also establishes robustness to the declared hanging-state reset noise. It does not establish model-error robustness, hardware transfer, minimum time, minimum force, or minimum rail.

7. Negative controls and lessons

The failed branches were scientifically useful because they localized each bottleneck.

- The unconditioned seven-link route scored 0/20. Exact trajectory quality did not imply a useful noisy launch basin.
- Passive waiting did not replace active hanging-state contraction.
- The eight-link open-loop route scored 0/20 with noise even after an exact replay succeeded.
- The 12 s, -0.20 m eight-link park scored 95/100 and was not promoted.
- Direct eight-to-nine transfer failed its initial probe.
- The balanced nine-link terminal objective left too much absolute-angle error for saturated LQR capture.
- The strongest early ten-link arrivals could be nearly upright at one sample while remaining far outside the five-second actuator-feasible capture set.
- Wider-rail, ghost-link, spring-support, CEM-tail, online-MPC, and static-gain probes were retained as development evidence but excluded from the releases.
- The seven-link controller failed tested changes in force authority, link mass

or length, damping, sensor noise, and delay. The released route is robust to its declared initial distribution, not arbitrary plant mismatch.

Three general lessons follow. State-coordinate schemas are part of the control contract. Terminal objectives must represent a dynamically quiet capture state. And empirical promotion must evaluate the complete switched controller, not an optimizer trace or a visually convincing crossing.

8. Public-record claim and literature boundary

8.1 Claim

As of 2026-09-14, we claim a provisional public link-count record for reset-free simulated cart-pole swing-up and sustained all-link upright hold under the benchmark frozen in Section 2: ten uniform passive serial links, one horizontal cart actuator, bounded force and rail, noisy hanging starts, and public multi-seed evidence.

The word `provisional` is essential. There is no recognized registry for this problem, and records across different plants are not automatically comparable. The claim is falsified by a prior public result meeting or exceeding ten links under the same or stricter declared dimensions, force convention, rail geometry, start distribution, action timing, sustained-upright rule, and reset policy.

8.2 Search performed

The public search used combinations of `n-link cart-pole`, `multi-link cart-pole swing-up`, `seven-link`, `ten-link`, `serial pendulum on cart`, `world record`, and `swing-up and stabilization`. The results were dominated by conventional single-pole benchmarks, double cart-poles, and `n-link` robots with actuators at multiple joints or only one passive joint. For example, Hirano et al. analyze an `n-link` planar robot with one passive revolute joint and demonstrate four links; that is a different actuation topology from an all-passive chain driven only by cart translation [6]. Standard continuous-control suites also focus on the single-pole cart-pole or double-pendulum variants [5,7]. No search result supplied a directly comparable seven-to-ten-link evidence bundle.

Search-engine absence is not proof of priority. Until independent comparison or peer review, the safest short public description is:

To our knowledge, this is the highest publicly evidenced link count for the precisely declared reset-free simulated cart-pole swing-up-and-hold benchmark.

8.3 What is not claimed

We do not claim a hardware record, a universal record across arbitrary force or rail limits, a minimum-time solution, an optimal controller, or a proof that no older unpublished solution exists. We also do not claim that one unchanged force trace solves all four plants. The shared contribution is the architecture and evidence discipline; the routes are target-plant solutions.

9. Reproduction and artifact map

9.1 Seven links

```
make setup-aligator
make release-swingup7
```

Primary manifest: `runs/swingup7_uniform/seven_link_swingup_manifest.json`

Video: `runs/swingup7_uniform/seven_link_swingup_success.mp4`

9.2 Eight links

```
PY=.conda-aligator/bin/python
$PY scripts/evaluate_fddp_parked_route.py \
  --config configs/swingup8_uniform.yaml \
  --controller runs/generalized_solver/n8_capture_fddp_feedback120.json \
  --episodes 100 --seed 80801 --park-seconds 14 --cart-target -0.15 \
  --tracking-gain-scale 0.75 \
  --out /tmp/n8_noisy100.json
```

Primary manifest: `runs/generalized_solver/eight_link_swingup_manifest.json`

Video: `runs/generalized_solver/eight_link_swingup_success.mp4`

9.3 Nine links

```
PY=.conda-aligator/bin/python
$PY scripts/evaluate_fddp_parked_route.py \
  --config configs/swingup9_uniform.yaml \
  --controller runs/generalized_solver/n9_fddp_terminal_angle20_widerail.json \
  --episodes 100 --seed 91041 --park-seconds 14 --cart-target -0.05 \
  --tracking-gain-scale 1.0 \
  --out /tmp/n9_noisy100.json
```

Primary manifest: `runs/generalized_solver/nine_link_swingup_manifest.json`

Video: `runs/generalized_solver/nine_link_swingup_success.mp4`

9.4 Ten links

```
PY=.conda-aligator/bin/python
$PY scripts/evaluate_fddp_parked_route.py \
  --config configs/swingup10_uniform.yaml \
  --controller runs/generalized_solver/n10_fddp_refined_route_feedback100.json \
  --episodes 100 --seed 102101 --park-seconds 16 --cart-target -0.05 \
  --tracking-gain-scale 1.0 --release-evidence \
  --out /tmp/n10_noisy100.json
```

```
PYTHONPATH=src $PY scripts/verify_ten_link_release.py
sha256sum -c runs/generalized_solver/n10_release_SHA256SUMS
```

Primary manifest: runs/generalized_solver/ten_link_swingup_manifest.json

Video: runs/generalized_solver/n10_fddp_refined_route_feedback100_park16_targetm005.mp4

The individual eight-, nine-, and ten-link notes remain in docs/ as detailed experiment appendices. The release manifests are authoritative if prose and a machine-readable field ever disagree.

10. Limitations and next work

The four results are simulated, exact-model, route-based controllers. The declared initial-state noise does not cover sensor noise during execution, actuator delay, unmodeled friction, parameter drift, or link manufacturing tolerances. No independent laboratory has yet reproduced the release. The state-faithful public videos are 2D renders of MuJoCo trajectories, not footage of a physical apparatus.

The conditioning phase is legitimate feedback control but consumes 10-16 s of the 30 s horizon. Faster controllers may exist. The routes are local solutions of a nonconvex trajectory problem and are not certificates of global optimality. The 100-episode gates characterize one frozen reset distribution; they are not formal reachability proofs.

The generalized-solver track should remain separate from the record release. It already expresses length, mass, force, damping, armature, timing, body geometry, and rail in dimensionless form; transfers route states and feedback by normalized arc length; and uses bounded exact-target continuation plus a thin actuator adapter. It has not yet automatically regenerated every 7-10 route from arbitrary unequal morphologies. That is the next methodological target.

11. Conclusion

A single bounded cart force can swing up and hold a ten-link passive serial chain from a noisy hanging start on a finite rail. The result was reached not by one monolithic learned policy, but by composing feedback funnels: contract the launch distribution at the stable equilibrium, track a target-plant nonlinear route, and enter the local upright controller only after the chain is dynamically quiet.

The progression from seven through ten links strengthens the claim because every promotion preserved the benchmark, control interface, reset rule, and evidence gate. It also clarifies what generalized: the architecture and experimental logic. The optimized swing route did not. The public artifacts support a provisional record claim under the exact frozen benchmark and invite the two tests that matter next: independent reproduction and a strictly matched prior-art comparison.

References

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